ADSICA

2025-04-10

```
# load the necessary package
library(dplyr)
library(readr)
library(tidyr)
library(tibble)
library(ggplot2)
library(tidyverse)
# read the data
df <- read_csv("2024-25-ads2-ica-life-expectancy.csv")
```

We checked the dataframe and we found that there exist some NA value in the columns of the CI, which may influence the process of handling the following problems.

```
# check the columns with NA
rows_with_na <- df[is.na(df$FactValueNumericLow), ]
# Check that the positions of missing values in both columns are identical
all(which(is.na(df$FactValueNumericLow)) ==
      which(is.na(df$FactValueNumericHigh)))
```

```
## [1] TRUE
```

We checked the rows with NA values and we thought there is no effects on the following process, so we decided to preserve these rows.

```
# data pre-processing
clean_df <- df %>%
  # select needed column
  select(Indicator = Indicator, Country = Location, Region = ParentLocation,
         Year = Period, Sex = Dim1, Life_Expectancy = FactValueNumeric,
         Lower_CI = FactValueNumericLow, Upper_CI = FactValueNumericHigh ) %>%
  # Standardized sex classification
  mutate(Indicator = case_when(
    Indicator == "Life expectancy at birth (years)" ~ "Birth",
    Indicator == "Life expectancy at age 60 (years)" ~ "60",
    TRUE ~ Indicator
  )) %>%
  mutate(Sex = case_when(
    Sex == "SEX_MLE" ~ "Male", Sex == "SEX_FMLE" ~ "Female",
    Sex == "SEX_BTSX" ~ "Both sexes",
    TRUE ~ Sex
  )) %>%
  # Standardized locale name
  mutate(Region = case_when(
    Region == "AFR" ~ "Africa", Region == "AMR" ~ "Americas",
    Region == "EMR" ~ "Eastern Mediterranean", Region == "EUR" ~ "Europe",
    Region == "SEAR" ~ "South-East Asia", Region == "WPR" ~ "Western Pacific",
    TRUE ~ Region)) %>%
  # Convert data type
```

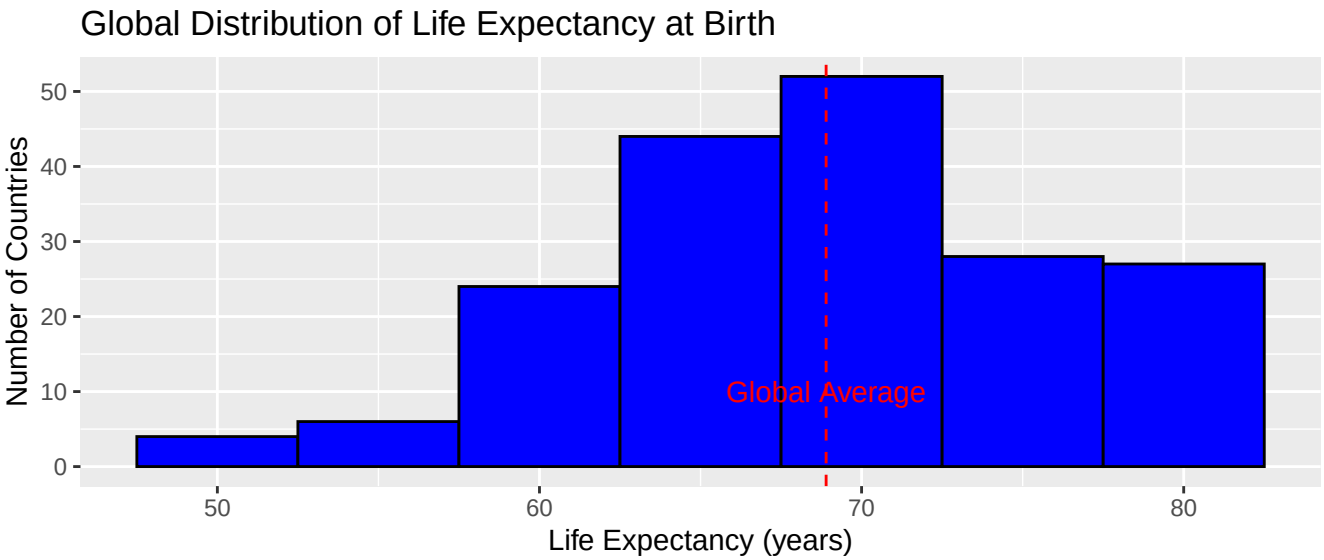
```
  mutate(
    Year = as.integer(Year),
    Life_Expectancy = round(Life_Expectancy, 2),
    Lower_CI = round(Lower_CI, 2), Upper_CI = round(Upper_CI, 2)) %>%
  # Add a confidence interval width
  mutate(CI_Width = Upper_CI - Lower_CI) %>%
  # ranking
  arrange(Region, Country, Year, Sex) %>%
  # Reset line number
  rowid_to_column("ID")
  clean_df$Life_Expectancy[clean_df$Indicator == 60] <-
  clean_df$Life_Expectancy[clean_df$Indicator == 60] + 60
```

1.1

```
life_expectancy <- clean_df
latest_year <- max(life_expectancy$Year, na.rm = TRUE)
life_expectancy_birth_latest <- life_expectancy %>%
  filter(Year == latest_year) %>%
  group_by(Country) %>%
  slice(which.min(Life_Expectancy)) %>%
  ungroup()
global_avg_life_expectancy <- life_expectancy_birth_latest %>%
  summarise(mean_life_expectancy = mean(Life_Expectancy, na.rm = TRUE))
global_avg_life_expectancy
```

```
## # A tibble: 1 x 1
##   mean_life_expectancy
##                  <dbl>
## 1                 68.9
```

```
ggplot(life_expectancy_birth_latest, aes(x = Life_Expectancy)) +
  geom_histogram(binwidth = 5, fill = "blue", color = "black") +
  labs(title = "Global Distribution of Life Expectancy at Birth",
       x = "Life Expectancy (years)",
       y = "Number of Countries") +
  geom_vline(xintercept = global_avg_life_expectancy$mean_life_expectancy,
             color = "red", linetype = "dashed") +
  annotate("text", x = global_avg_life_expectancy$mean_life_expectancy,
           y = 10, label = "Global Average", color = "red")
```

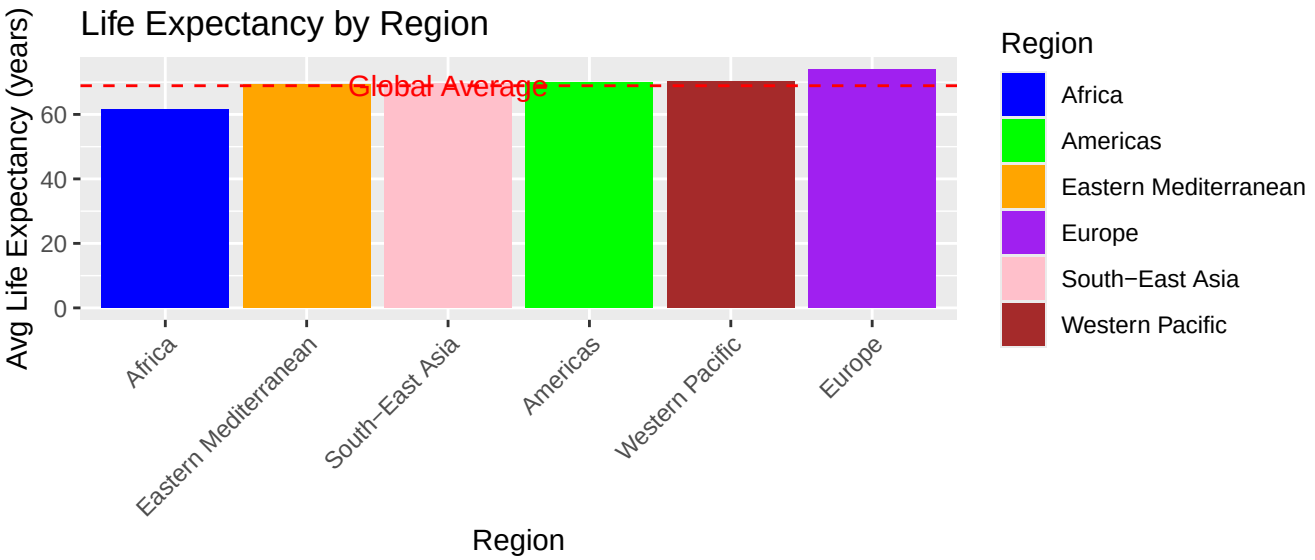


The average life expectancy at birth worldwide is around 17.25, most countries' life expectancies at birth worldwide are around this figure.

```
region_avg_life_expectancy <- life_expectancy_birth_latest %>%
  group_by(Region) %>%
  summarise(mean_life_expectancy = mean(Life_Expectancy, na.rm = TRUE))
regions_above_avg <- region_avg_life_expectancy %>%
  filter(mean_life_expectancy > global_avg_life_expectancy$mean_life_expectancy)
regions_above_avg
```

```
## # A tibble: 5 x 2
##   Region                mean_life_expectancy
##   <chr>                                <dbl>
## 1 Americas                              70.1
## 2 Eastern Mediterranean                 69.4
## 3 Europe                                74.1
## 4 South-East Asia                       69.8
## 5 Western Pacific                       70.1
```

```
life_expectancy_birth_latest <- life_expectancy %>% filter(Year == latest_year) %>%
  group_by(Region, Country) %>% slice(which.min(Life_Expectancy)) %>% ungroup()
region_avg_life_expectancy <- life_expectancy_birth_latest %>% group_by(Region) %>%
  summarise(mean_life_expectancy = mean(Life_Expectancy, na.rm = TRUE))
global_avg_life_expectancy <- life_expectancy_birth_latest %>%
  summarise(mean_life_expectancy = mean(Life_Expectancy, na.rm = TRUE))
regions_above_avg <- region_avg_life_expectancy %>%
  filter(mean_life_expectancy > global_avg_life_expectancy$mean_life_expectancy)
ggplot(region_avg_life_expectancy, aes(x = reorder(Region, mean_life_expectancy),
                                       y = mean_life_expectancy, fill = Region)) +
  geom_bar(stat = "identity") + labs(title = "Life Expectancy by Region",
                                     x = "Region", y = "Avg Life Expectancy (years)") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  geom_hline(yintercept = global_avg_life_expectancy$mean_life_expectancy,
             color = "red", linetype = "dashed") +
  annotate("text", x = 3, y = global_avg_life_expectancy$mean_life_expectancy,
           label = "Global Average", color = "red") +
  scale_fill_manual(
    values = c("blue", "green", "orange", "purple", "pink", "brown"),
    labels = c("Africa", "Americas", "Eastern Mediterranean",
               "Europe", "South-East Asia", "Western Pacific"), guide = "legend")
```



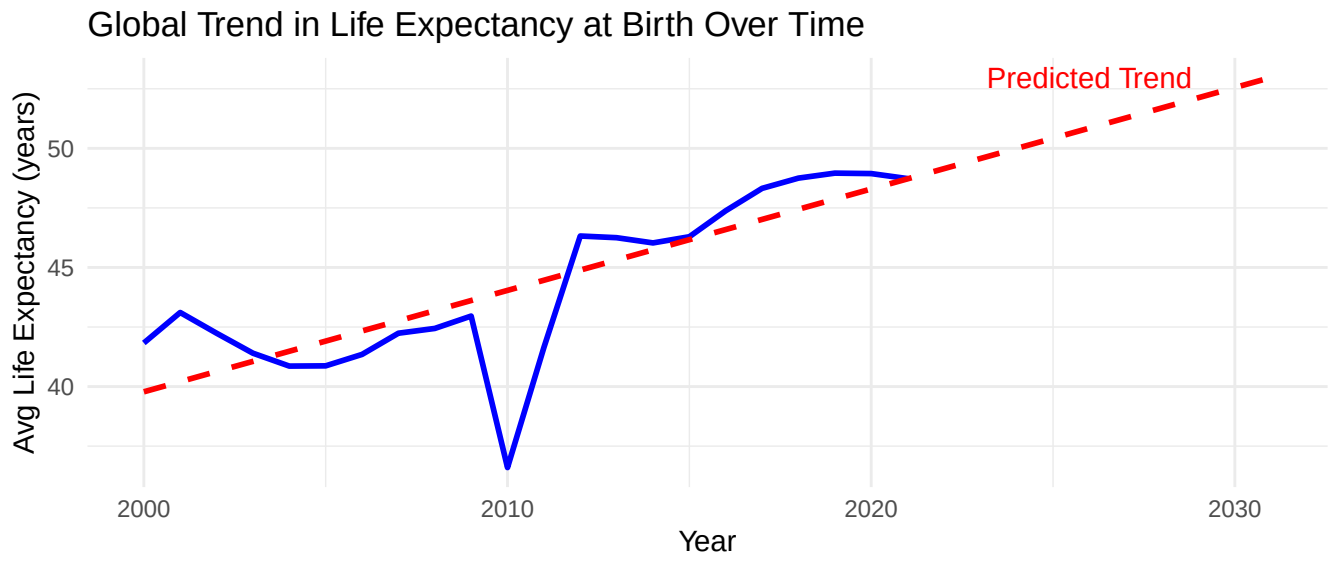
In Africa, the life expectancy at birth is the lowest at 15 years, while the Western Pacific region has the highest at 17.8 years. Four regions have life expectancy at birth above the average: the Americas, Europe, South-East Asia, and the Western Pacific.

```
latest_year <- max(life_expectancy$Year, na.rm = TRUE)
life_expectancy_global_avg <- life_expectancy %>%
  group_by(Year) %>% slice(which.min(Life_Expectancy)) %>%
  summarise(Global_Life_Expectancy = mean(Life_Expectancy, na.rm = TRUE)) %>%
  ungroup()
life_expectancy_model <- lm(Global_Life_Expectancy ~ Year, data =
                              life_expectancy_global_avg)
current_year <- max(life_expectancy_global_avg$Year, na.rm = TRUE)
future_year <- current_year + 5
predicted_life_expectancy <- predict(life_expectancy_model, newdata =
                                       data.frame(Year = future_year))
predicted_life_expectancy_value <- as.numeric(predicted_life_expectancy)
predicted_life_expectancy_value
```

```
## [1] 50.84876
```

```
life_expectancy <- life_expectancy %>%
  mutate(Year = as.numeric(Year))
life_expectancy_min <- life_expectancy %>% group_by(Year) %>%
  summarise(min_Life_Expectancy = min(Life_Expectancy, na.rm = TRUE)) %>%
  ungroup()
global_trend <- life_expectancy_min %>% group_by(Year) %>%
  summarise(global_life_expectancy = mean(min_Life_Expectancy, na.rm = TRUE)) %>%
  ungroup()
life_expectancy_model <- lm(global_life_expectancy ~ Year, data = global_trend)
future_years <- seq(min(global_trend$Year), max(global_trend$Year) + 10, by = 1)
predicted_values <- predict(life_expectancy_model, newdata =
                              data.frame(Year = future_years))
global_trend_predicted <- data.frame(Year = future_years,
                                     global_life_expectancy = predicted_values)
ggplot() +
  geom_line(data = global_trend, aes(x = Year, y = global_life_expectancy),
            color = "blue", linewidth = 1) +
  geom_line(data = global_trend_predicted,
            aes(x = Year, y = global_life_expectancy),
            color = "red", linetype = "dashed", size = 1) +
```

```
  labs(title = "Global Trend in Life Expectancy at Birth Over Time",
       x = "Year", y = "Avg Life Expectancy (years)") + theme_minimal() +
  annotate("text", x = max(global_trend$Year) + 5, y = max(predicted_values),
           label = "Predicted Trend", color = "red")
```



Assuming current trends do not change, the life expectancy 5 years later is around 10.72. From the chart, it can be seen that the overall trend is upward. Although there has been a recent downward trend, there have been instances of decreases followed by increases in the past. Therefore, it is predicted that the value five years from now will be greater than the current one.

1.2 Country that has the highest life expectancy at birth(latest year)

```
cleaned_data= clean_df
birth_data = cleaned_data[cleaned_data$Indicator == "Birth", ]
latest_year = max(cleaned_data$Year)
latest_year_birth = birth_data[birth_data$Year == latest_year, ]
highest_life_expectancy = max(latest_year_birth$Life_Expectancy)
highest_country_data= latest_year_birth [latest_year_birth$Life_Expectancy
                                         == highest_life_expectancy, ]
highest_country = highest_country_data$Country
highest_country_data
```

```
## # A tibble: 1 x 10
##      ID Indicator Country Region    Year Sex   Life_Expectancy Lower_CI Upper_CI
##   <int> <chr>     <chr>   <chr>    <int> <chr>           <dbl>    <dbl>    <dbl>
## 1 22437 Birth     Japan   Western~  2021 Fema~            87.2     87.2     87.2
## # i 1 more variable: CI_Width <dbl>
```

The country that has the highest life expectancy at birth according to the latest year (2021) of this dataset is FEMALE in JAPAN.

The trend of life expectancy at birth in that country

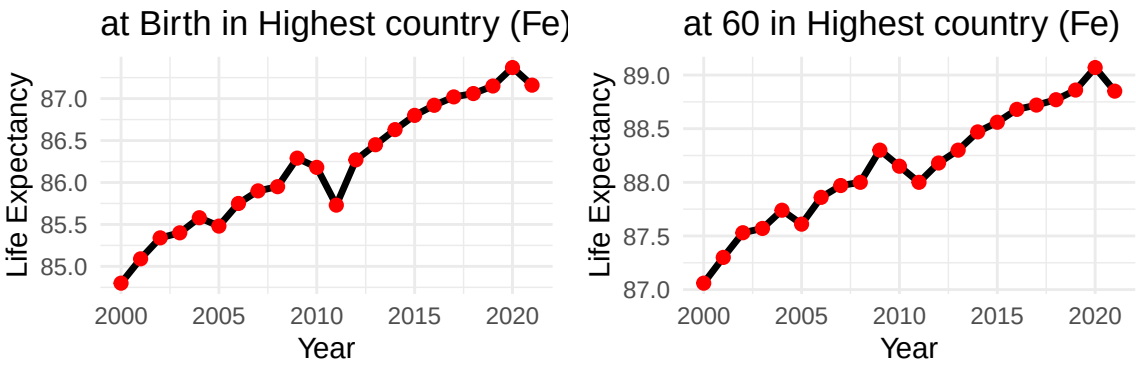
```
highest_country_female_birth_trend = birth_data [birth_data$Country ==
                                                   highest_country
                                                 & birth_data$Sex == "Female", ]
birth_female_plot = ggplot(highest_country_female_birth_trend,
                    aes(x = Year, y = Life_Expectancy)) + geom_line(size = 1.2) +
                    geom_point (size = 2, color = "red")+
  labs(title = "at Birth in Highest country (Fe)",
       y = "Life Expectancy", x = "Year") +
  theme_minimal()
```

Compare the life expectancy trends of both indicators “Birth” and “60” in the country with the highest life expectancy

```
highest_country_age60_trend = cleaned_data[cleaned_data$Country ==
                                             highest_country
                                           & cleaned_data$Indicator == "60"
birth_age60_female_plot = ggplot(highest_country_age60_trend,
            aes(x = Year, y = Life_Expectancy)) + geom_line (size = 1.2) +
            geom_point (size = 2, color = "red")+
  labs (title = "at 60 in Highest country (Fe)",
        y = "Life Expectancy", x = "Year") +
  theme_minimal()
```

To improve readability and comparison, both trends are presented side by side

```
birth_female_plot
birth_age60_female_plot
```

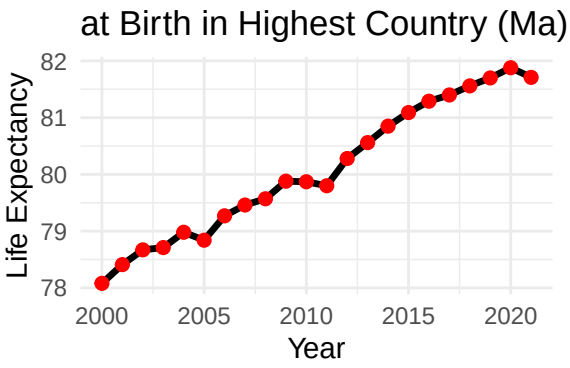


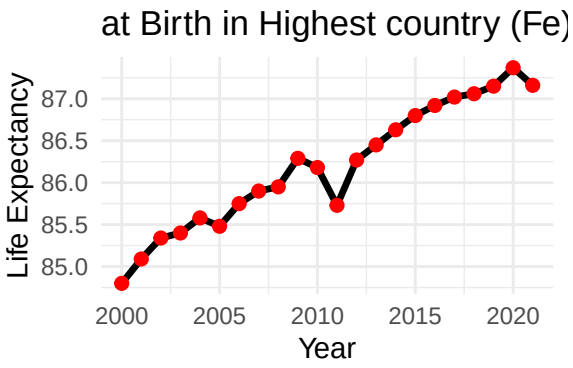
Based on the data provided and the graphs, the life expectancy at birth and at age 60 show similar shape and positive correlation in Japan from year 2000 to year 2021. Around Year 2011-2012 and Year 2020-2021, the decrease of life expectancy in both indicators may due to some national health impacts such as earthquake and COVID-19.

Compare the life expectancy trends of male and female in the country with the highest life expectancy

```
highest_country_male_birth = birth_data[birth_data$Sex == "Male"
                                        & birth_data$Country == highest_country, ]
highest_country_male_birth_plot =ggplot (highest_country_male_birth,
          aes(x = Year, y = Life_Expectancy))+
          geom_line (size = 1.2)+ geom_point (size = 2, color = "red")+
  labs (title = "at Birth in Highest Country (Ma)",
        y = "Life Expectancy",
        x = "Year")+
  theme_minimal()
```

```
highest_country_male_birth_plot
birth_female_plot
```





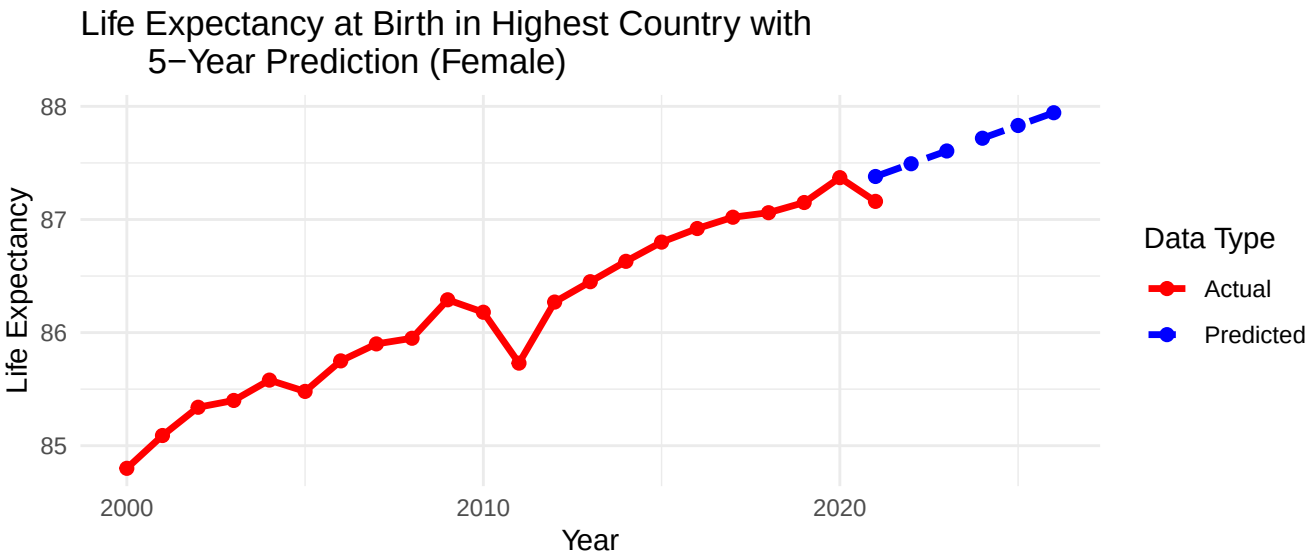
Based on the data provided and the graphs, the female and male life expectancy show similar shape and positive correlation in Japan from year 2000 to year 2021. However, women consistently live longer than man in Japan(~6 years longer). While women start from a higher baseline, the rate of improvement appears faster for males. The higher baseline in female life expectancy could explain the sharper decline observed around Year 2011 and 2012. For instance, elderly women, who make up a large portion of Japan's aging group, may be more vulnerable during disasters due to the fragile health, leading to a more noticeable drop compared to the male.

Assuming current trends continue, prediction of life expectancy 5 years later in the country with the highest life expectancy

```
model = lm(Life_Expectancy ~ Year, data = highest_country_female_birth_trend)
latest_year = max(highest_country_female_birth_trend$Year)
future_years = data.frame(Year = seq(latest_year, latest_year + 5))
future_years$Life_Expectancy = predict(model, newdata = future_years)
future_years$Sex = "Female"
future_years$Indicator = "Birth"
combined_data =rbind(
  highest_country_female_birth_trend[, c("Year", "Life_Expectancy",
                                         "Sex", "Indicator")], future_years)
tail(combined_data, 5)
```

```
## # A tibble: 5 x 4
##    Year Life_Expectancy Sex    Indicator
##   <int>           <dbl> <chr>  <chr>
## 1  2022            87.5 Female Birth
## 2  2023            87.6 Female Birth
## 3  2024            87.7 Female Birth
## 4  2025            87.8 Female Birth
## 5  2026            87.9 Female Birth
```

```
ggplot() +
  geom_line(data = highest_country_female_birth_trend,
            aes(x = Year, y = Life_Expectancy, color = "Actual"), size = 1.2) +
  geom_point(data = highest_country_female_birth_trend,
            aes(x = Year, y = Life_Expectancy, color = "Actual"), size = 2) +
  geom_line(data = future_years,
            aes(x = Year, y = Life_Expectancy, color = "Predicted"),
            size = 1.2, linetype = "dashed") +
  geom_point(data = future_years,
            aes(x = Year, y = Life_Expectancy, color = "Predicted"), size = 2) +
  scale_color_manual(name = "Data Type",
                     values = c("Actual" = "red", "Predicted" = "blue")) +
  labs(title = "Life Expectancy at Birth in Highest Country with
       5-Year Prediction (Female)", x = "Year", y = "Life Expectancy") +
  theme_minimal()
```



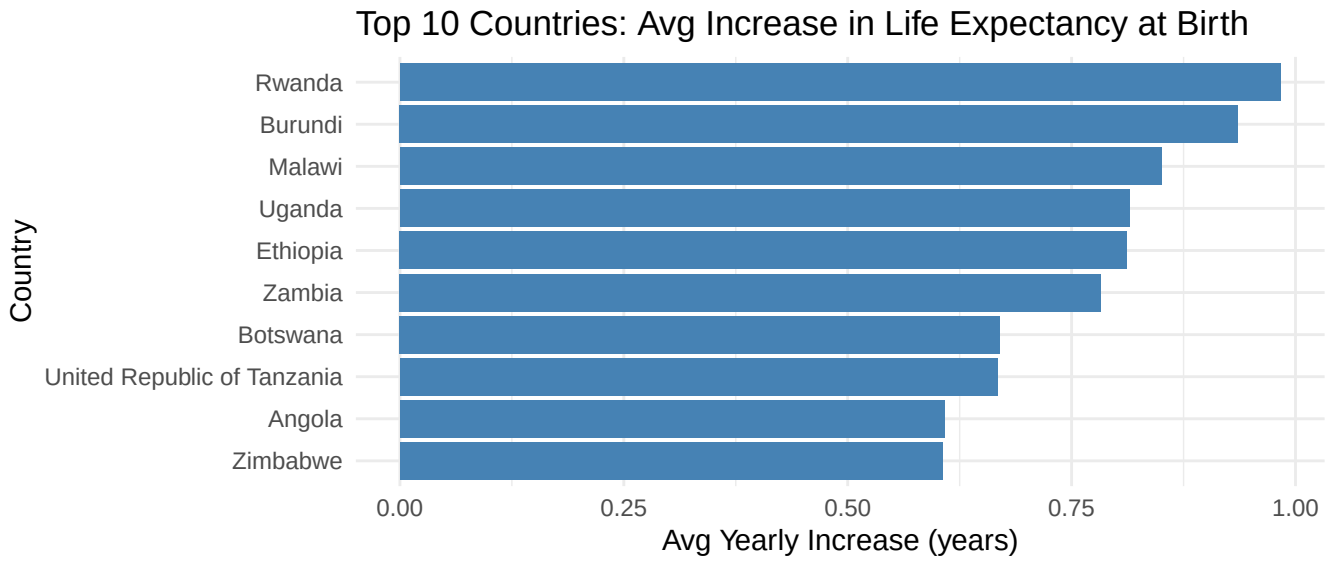
1.3 Question 3: Top 10 Countries with Highest Average Increase in Life Expectancy Read cleaned data

Life Expectancy at Birth: Top 10 Countries by Average Yearly Increase

```
birth_life <- clean_df %>%
  filter(Indicator == "Birth", Sex == "Both sexes") %>% group_by(Country) %>%
  arrange(Year) %>%
  # Calculate year-to-year change
  mutate(change = Life_Expectancy - lag(Life_Expectancy)) %>%
  # Compute average yearly change per country
  summarise(avg_change_birth = mean(change, na.rm = TRUE), n_years = n()) %>%
  arrange(desc(avg_change_birth)) %>%
  # Get top 10 countries
  slice_head(n = 10)
```

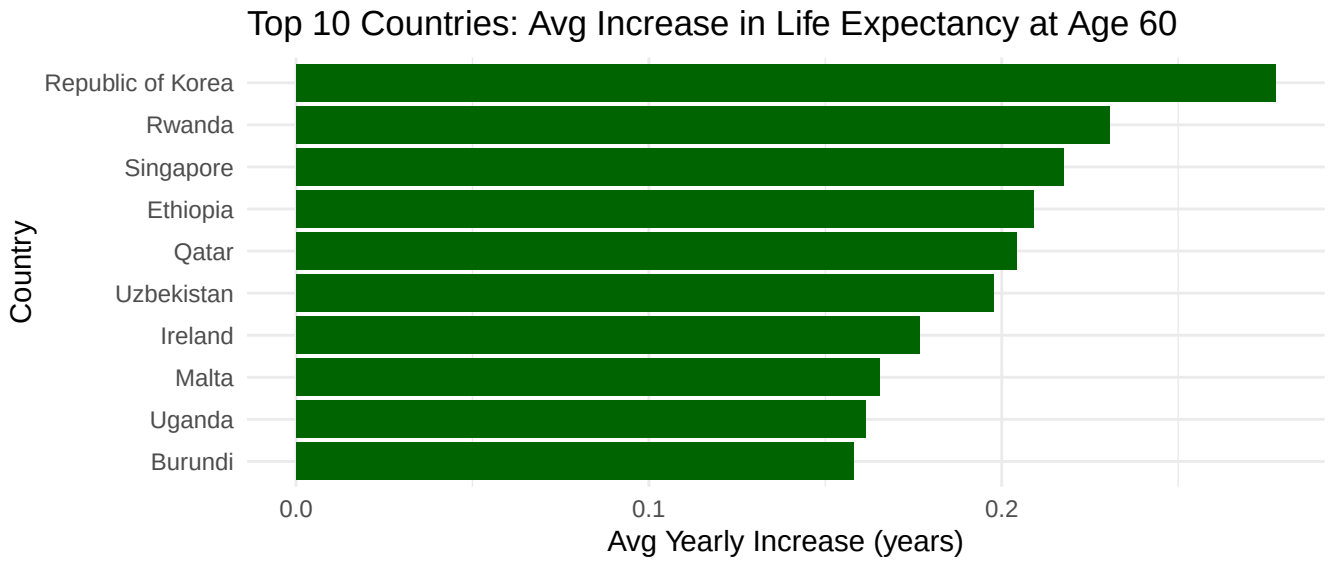
Visualization: Avg Increase in Life Expectancy at Birth (Top 10 Countries)

```
library(ggplot2)
ggplot(birth_life, aes(x = reorder(Country, avg_change_birth),
                       y = avg_change_birth)) +
  geom_col(fill = "steelblue") + coord_flip() +
  labs(title = "Top 10 Countries: Avg Increase in Life Expectancy at Birth",
       x = "Country", y = "Avg Yearly Increase (years)") +
  theme_minimal()
```



Life Expectancy at Age 60: Top 10 Countries by Average Yearly Increase and visualization

```
# Filter data for life expectancy at age 60 and both sexes
life_at_60 <- clean_df %>%
  filter(Indicator == "60", Sex == "Both sexes") %>%
  group_by(Country) %>%
  arrange(Year) %>%
  # Calculate year-to-year change
  mutate(change = Life_Expectancy - lag(Life_Expectancy)) %>%
  # Compute average yearly change per country
  summarise(avg_change_60 = mean(change, na.rm = TRUE), n_years = n()) %>%
  arrange(desc(avg_change_60)) %>%
  # Get top 10 countries
  slice_head(n = 10)
ggplot(life_at_60, aes(x = reorder(Country, avg_change_60),
                       y = avg_change_60)) +
  geom_col(fill = "darkgreen") +
  coord_flip() +
  labs(title = "Top 10 Countries: Avg Increase in Life Expectancy at Age 60",
       x = "Country", y = "Avg Yearly Increase (years)") +
  theme_minimal()
```



The countries showing the highest gains in life expectancy at birth are not the same as those with the highest gains at age 60. This highlights how different health policies and improvements affect different age groups. Birth life expectancy gains reflect improvements in early-life and basic health systems, while gains at age 60 reflect long-term healthcare effectiveness and elderly support.

2.1

Task 2

Whether economic conditions could explain variations in life expectancy across countries?

We wanted to explore whether overall economic conditions is a key factor affecting on life expectancy throughout ages, so we checked GDP per capita(PPP) across countries from 2000 to 2021.

We chose data from World Bank Open Data, the indicator "NY.GDP.PCAP.PP.KD" implies the GDP per capita (PPP) adjusted for purchasing power parity, which eliminates price differentials and can more accurately compare real living standards across countries.

```
library(WDI)
```

```
gdp_ppp <- WDI(
  indicator = "NY.GDP.PCAP.PP.KD",
  start = 2000,
  end = 2021)
head(gdp_ppp,3)
```

```
##                       country iso2c iso3c year NY.GDP.PCAP.PP.KD
## 1 Africa Eastern and Southern    ZH   AFE 2021          3934.287
## 2 Africa Eastern and Southern    ZH   AFE 2020          3861.111
## 3 Africa Eastern and Southern    ZH   AFE 2019          4073.881
```

Firstly we wanted to compare the average gdp per capita with average life expectancy, regardless of the time and sex.

We fetch df only by the location, the location type and FactValueNumeric.

```
df_t2 <- df %>%
  # select needed column
  select(
    Location = Location,
    Life_Expectancy = FactValueNumeric) %>%
  arrange(Location, Life_Expectancy)
head(df_t2,3)
```

```
## # A tibble: 3 x 2
##   Location    Life_Expectancy
##   <chr>                 <dbl>
## 1 Afghanistan            13.1
## 2 Afghanistan            13.1
## 3 Afghanistan            13.1
```

Now let's deal with the gdp_ppp data frame. The "country" column contains countries and regions, so we only save the country and GDP.

```
gdp_ppp <- gdp_ppp %>%
  select(country = country,
         GDP_PPP = NY.GDP.PCAP.PP.KD)
head(gdp_ppp,3)
```

```
##                       country  GDP_PPP
## 1 Africa Eastern and Southern 3934.287
## 2 Africa Eastern and Southern 3861.111
## 3 Africa Eastern and Southern 4073.881
```

We matched the country between the two dataframes. For the countries and areas contained in df_t2, we saved the corresponding lines in gdp_ppp.

```
matched_gdp <- gdp_ppp %>%
  filter(trimws(tolower(country)) %in%
      trimws(tolower(df_t2$Location)))
head(matched_gdp,3)
```

```
##       country  GDP_PPP
## 1 Afghanistan 2144.167
## 2 Afghanistan 2769.686
## 3 Afghanistan 2927.245
```

```
#check whether there is location in df_t2 unmatched
unmatched <- df_t2 %>%
  anti_join(gdp_ppp, by = c("Location" = "country")) %>%
  distinct(Location)
print(paste("number of countries unmatched:", nrow(unmatched)))
```

```
## [1] "number of countries unmatched: 24"
```

There is too many unmatched countries. We discovered that there are some names differently expressed in both tables, and we matched them by hands.

```
#checking which names are different, the list is too long so we hide it.
print(unmatched)
unmatched2 <- gdp_ppp %>%
  anti_join(df_t2, by = c("country" = "Location")) %>%
  distinct(country)
print(unmatched2)
```

```
#adjusting the country name in gdp_ppp
gdp_ppp <- gdp_ppp%>%
  mutate(country=case_when(
    country == "Bahamas, The" ~ "Bahamas",
    country == "Bolivia" ~ "Bolivia (Plurinational State of)",
    country == "Congo, Rep." ~ "Congo",
    country == "Korea, Dem. People's Rep." ~
      "Democratic People's Republic of Korea",
    country == "Congo, Dem. Rep." ~ "Democratic Republic of the Congo",
    country == "Egypt, Arab Rep." ~ "Egypt",
    country == "Gambia, The" ~ "Gambia",
    country == "Iran, Islamic Rep." ~ "Iran (Islamic Republic of)",
    country == "Lao PDR" ~ "Lao People's Democratic Republic",
    country == "Micronesia, Fed. Sts." ~ "Micronesia (Federated States of)",
    country == "Netherlands" ~ "Netherlands (Kingdom of the)",
    country == "Korea, Rep." ~ "Republic of Korea",
    country == "Kyrgyz Republic" ~ "Kyrgyzstan",
    country == "Moldova" ~ "Republic of Moldova",
    country == "St. Lucia" ~ "Saint Lucia",
    country == "St. Vincent and the Grenadines"
    ~ "Saint Vincent and the Grenadines",
    country == "Slovak Republic" ~ "Slovakia",
    country == "Turkiye" ~ "Türkiye",
    country == "United Kingdom"
    ~ "United Kingdom of Great Britain and Northern Ireland",
    country == "Tanzania" ~ "United Republic of Tanzania",
```

```
    country == "United States" ~ "United States of America",
    country == "Venezuela, RB" ~ "Venezuela (Bolivarian Republic of)",
    country == "Yemen, Rep." ~ "Yemen",
    TRUE ~ country
  ))
```

```
matched_gdp_2 <- gdp_ppp %>%
  filter(trimws(tolower(country)) %in%
      trimws(tolower(df_t2$Location)))
unmatched_after <- df_t2 %>%
  anti_join(gdp_ppp, by = c("Location" = "country")) %>%
  distinct(Location)
print(paste("number of countries unmatched:", nrow(unmatched_after)))
```

```
## [1] "number of countries unmatched: 1"
```

Now the matching rate is higher. The only one not matched "occupied Palestinian territory, including east Jerusalem" don't affect the whole scene, and we just delete it.

Secondly, we caculated the mean and merged two tables.

```
#Caculate mean of life expectancy by country
mean_df <- df_t2 %>%
  select(country = Location,
         Life_Expectancy = Life_Expectancy) %>%
  group_by(country) %>%
  summarise(avg_life = mean(Life_Expectancy, na.rm = TRUE),.groups = "drop")
#Caculate mean of gdp_ppp by country
mean_gdp <- matched_gdp_2 %>%
  select(country = country,
         GDP_PPP = GDP_PPP) %>%
  group_by(country) %>%
  summarise(avg_ppp = mean(GDP_PPP, na.rm = TRUE),.groups = "drop")
merge_df <- merge(mean_df,mean_gdp)
merge_df <- merge_df %>% filter(!is.na(avg_ppp))
merge_df <- merge_df %>% filter(!is.na(avg_life))
head(merge_df,3)
```

```
##       country avg_life  avg_ppp
## 1 Afghanistan 36.32424  2430.33
## 2     Albania 48.49485 11416.89
## 3     Algeria 47.88121 14125.52
```

GDP data typically show a right-skewed distribution (positive skew) because a few high-income countries have much larger GDP than most countries, resulting in a concentration of data on the left with a long tail of extremely large values on the right. So we did log transformation to reduce skewness and linearize relationships.

```
merge_df <- merge_df %>%
  mutate(log_ppp = log(avg_ppp))
head(merge_df,3)
```

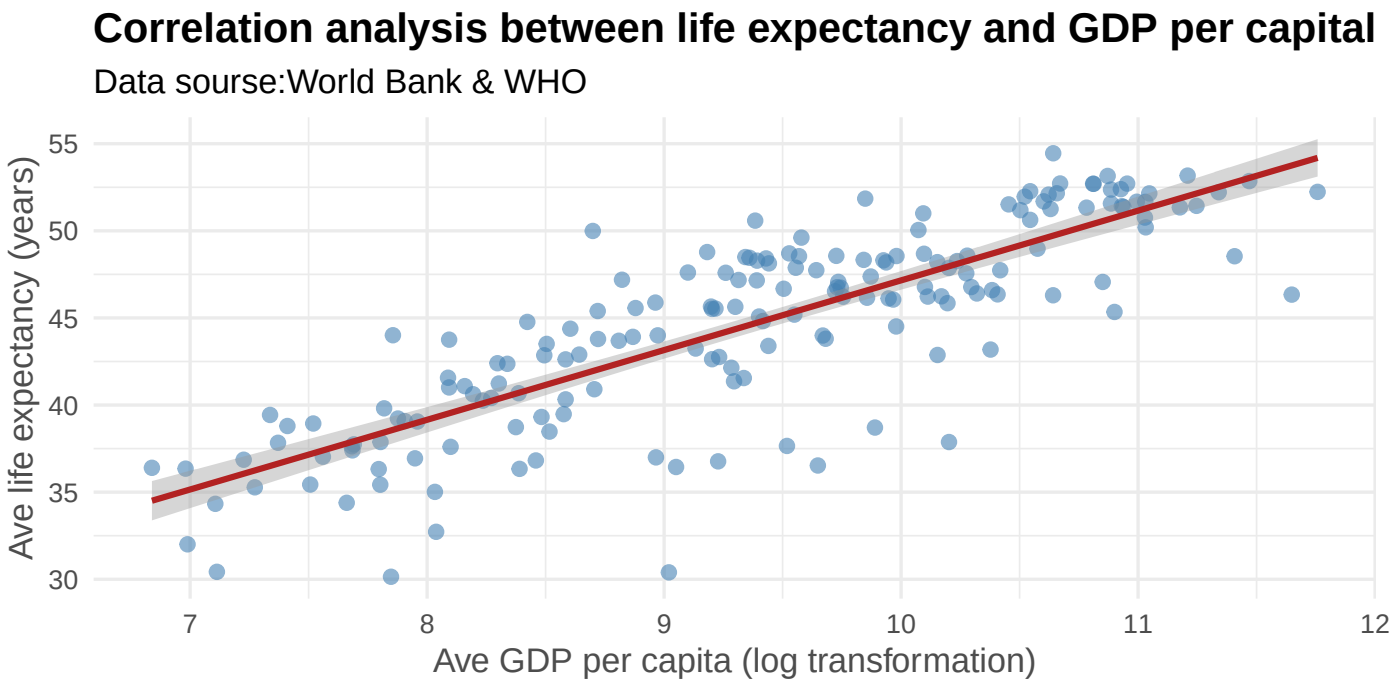
```
##       country avg_life  avg_ppp  log_ppp
## 1 Afghanistan 36.32424  2430.33 7.795782
## 2     Albania 48.49485 11416.89 9.342849
## 3     Algeria 47.88121 14125.52 9.555738
```

Scatter plot with correlation analysis about average life expectancy and gdp per capita

```
cor_matrix <- cor(merge_df$avg_life, merge_df$log_ppp, method = "pearson")
cat("Pearson correlation coefficient:", round(cor_matrix, 2))
```

```
## Pearson correlation coefficient: 0.83
```

```
ggplot(merge_df, aes(x = log_ppp, y = avg_life)) +
  geom_point(alpha = 0.6, color = "steelblue", size = 2.5) +
  geom_smooth(method = "lm", formula = y ~ x, color = "firebrick",
              se = TRUE, linewidth = 1.2) +
  labs(
    title = "Correlation analysis between life expectancy and GDP per capital",
    subtitle = "Data sourse: World Bank & WHO",
    x = "Ave GDP per capita (log transformation)",
    y = "Ave life expectancy (years)") +
  theme_minimal(base_size = 14) +
  theme(plot.title = element_text(face = "bold", hjust = 0.5),
    axis.title = element_text(color = "grey30"))
```



With the Pearson coefficient 0.83, we can get to conclusion that there is a rough linear relationship between life expectancy and GDP per capit across countries. To further ensure it we made a linear regression model

```
model <- lm(avg_life ~ log_ppp, data = merge_df)
summary(model)
```

```
##
## Call:
## lm(formula = avg_life ~ log_ppp, data = merge_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.8386  -1.5560   0.5324   2.0120   8.0413
##
## Coefficients:
```

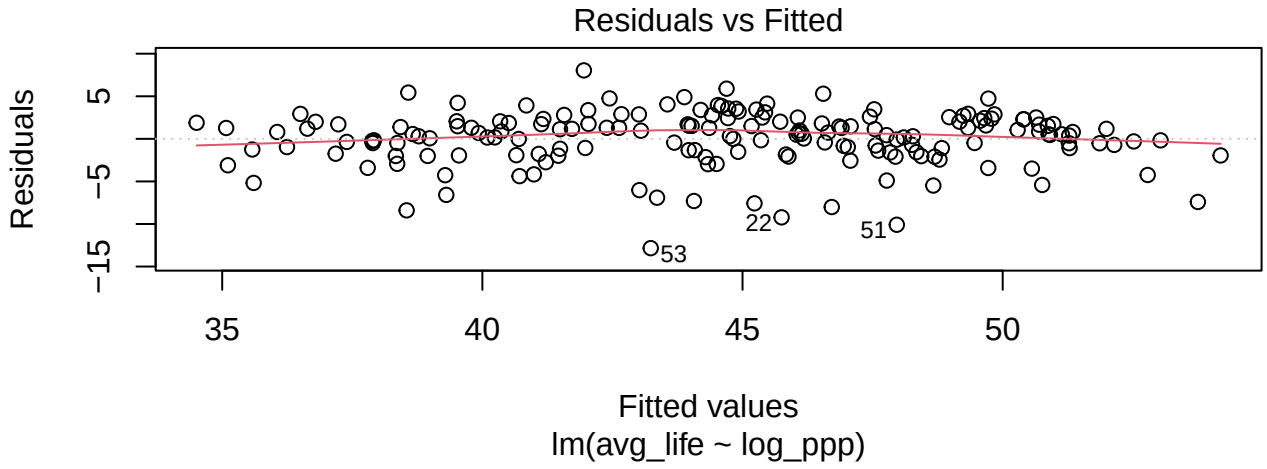
```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.1463     1.9388   3.686 0.000303 ***
## log_ppp       4.0010     0.2053  19.492  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.181 on 176 degrees of freedom
## Multiple R-squared:  0.6834, Adjusted R-squared:  0.6816
## F-statistic: 379.9 on 1 and 176 DF,  p-value: < 2.2e-16
```

```
coefficients(model)
```

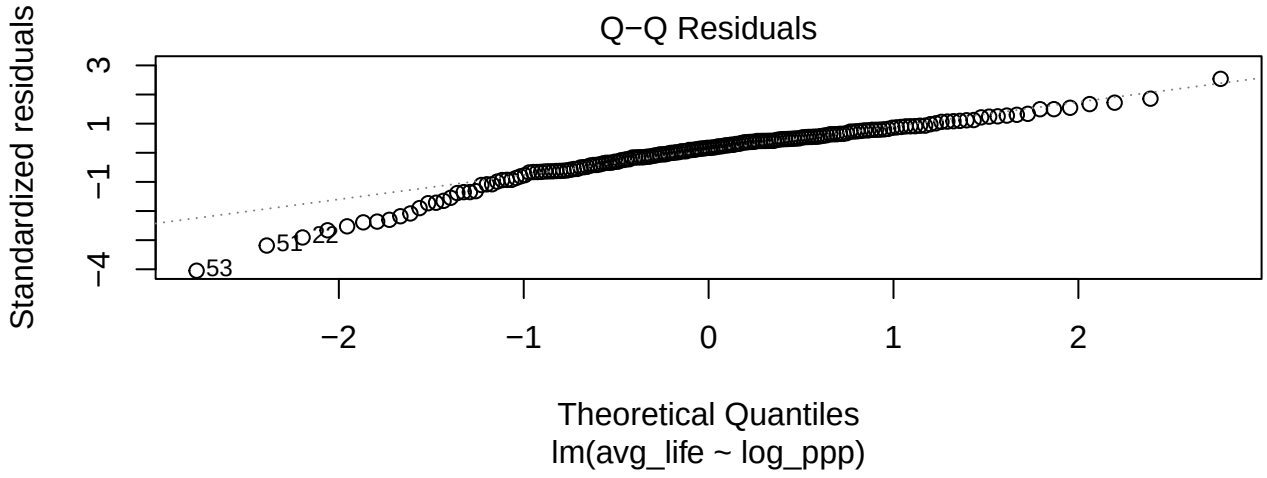
```
## (Intercept)     log_ppp
##    7.146310    4.000969
```

Then we draw residual plot and qq plot to examine the assumptions of linear regression model.

```
plot(model,1)
```




```
plot(model,2)
```



From above we can know that the residuals are roughly normally distributed and homoscedastic.

Understanding of our Result

We only get a rough, overall trend of both the life expectancy and the GDP per capital for each countries, and proved that there's a linear relationship.

However, we only used GDP per capita to describe economic levels, and we ignored different times and sexes of samples. Therefore, for a further analysis, maybe we can look into different sexes, different kind of economic indicators(eg.The proportion of pension expenditure for samples of age 60), etc.

What's more, having linear relationship doen't strictly means that GDP can explain the distribution of life expectancy. To further prove this more background information is needed.